Pre-Analysis Plan: ManyDaughters

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1 Hypotheses:

H1: Having daughters leads to stronger support for gender equality (as measured by attitudes).
H2: Having daughters leads to more left-leaning political preferences.
H3: Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).
H4: Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

2 Variables

Our unit of analysis will be the dyad (i.e., the couple). For each couple, we will define the following variables. Note that participants will be matched into dyads using the variable "parid" from the "ppathl" dataset.

Dependent variables:

- Support for gender equality (attitudes): We will derive a composite measure of support for gender equality using the following variables from the "pl" dataset: "plj0621", "plj0622", "plj0623", "plj0624", and "plj0625". These variables capture respondents' agreement ratings (from 1 – do not agree at all, to 7 – totally agree) with the following statements: "Having a job is the best way for a woman to be independent", "Even a married woman should have a paid job so that she can be financially independent", "If a woman earns more money than her partner, this inevitably leads to problems", "For parents, vocational training or higher education for their sons should be more important than vocational training or higher education for their daughters", "At home, the husband should have the final say". To aggregate these responses for each individual in the dyad, we will employ principal component analysis (PCA). First, we will reverse-score the last three variables ("plj0623", "plj0624", "plj0625") so that higher ratings reflect greater support for gender equality. Then, we will standardize all five variables. Next, we will conduct

PCA and extract the first principal component. This procedure will yield a single attitudinal measure per individual representing support for gender equality. We will then aggregate this measure at the dyad level by averaging the values for the two individuals in the couple.

- Left-leaning political preferences: The variable "plh0004" from the "pl" dataset records participants' responses to the following question: "In politics people sometimes talk of left and right. Where would you place your self on a scale from 0 to 10 where 0 means the left and 10 means the right?". We will reverse-code this variable so that higher values indicate more left-leaning political preferences. We will then aggregate this measure at the dyad level (for a given survey year) by averaging the values for the two individuals in the couple.

- Gender equality (behaviors): We will derive a behavioral measure of gender equality by taking the following steps. First, for each individual, we will compute the weekly sum of hours devoted to housework and childcare. To do so, we will sum (for a given individual and survey year) the following variables from the "pl" dataset to estimate the number of hours devoted to these activities on weekdays: "pli0043_h" and "pli0044_h". We will apply the same procedure using "pli0012_h" and "pli0019_h" for Saturdays, and "pli0016_h" and "pli0022_h" for Sundays. Using these three measures, we will estimate the total number of hours devoted weekly (per individual and survey year) to housework and childcare. Next, for each dyad, we will calculate the deviation from an equal (50/50) division of time spent on these activities between partners. Specifically, we will compute the absolute difference in time devoted by the two individuals and divide it by their total combined time. This results in a measure ranging from 0 to 1, where higher values indicate greater inequality. To obtain our final measure of gender equality in the dyad, we will subtract this inequality score from 1. The result will be a single measure per dyad and year, with higher values reflecting more gender equality in the time spent on these activities.

- Differences in support for gender equality (attitudes): We will obtain a measure of the attitudinal gap in support for gender equality within each dyad. As described in the definition of our first dependent variable, we computed an individual-level index of support for gender equality by applying PCA to the variables "plj0621", "plj0622", "plj0623", "plj0624", and "plj0625". Instead of averaging these scores across the couple (as we did for our first attitudinal measure), we will now calculate the difference in attitudes by subtracting the PCA-derived score of the female partner from that of the male partner. This resulting value captures the within-dyad difference in support for gender equality.

Main independent variable:

- Daughter: This variable takes the value 1 if the couple had at least one daughter born in or before the survey year, and 0 otherwise. To construct this variable, we will use information from the "biobirth" dataset, which includes the year of birth and gender of each child, recorded in the variables "kidgeb01", "kidgeb02", ..., "kidgeb19" and "kidsex01", "kidsex02", ..., "kidsex19". For a given dyad and survey year, we will loop over all recorded children and check two conditions: (1) the child's year of birth (kidgebXX) is less than or equal to the survey year (syear), and (2) the child's gender (kidsexXX) is coded as 2 (indicating female). If at least one child meets both conditions, this variable will be coded as 1 for that dyad. Otherwise, it will be coded as 0.

In our analyses, we will include a set of demographic control variables to account for potential confounders. For a given dyad and survey year, these include:

- Average age and average age squared for the couple: Calculated using the year of birth ("gebjahr") from the "ppathl" dataset and the survey year.
- Employment status (female partner), employment status (male partner): Derived from the variable "pgemplst" in the "pgen" dataset.
- Highest school-leaving degree (female partner), highest school-leaving degree (male partner): Based on the variable "pgpsbil" from the "pgen" dataset.
- Federal state indicators: A set of binary variables representing the 16 German federal states, constructed using the variable "hgnuts1" from the "hgen" dataset.
- Log of yearly net household income: Derived from the variable "hghinc" in the "hgen" dataset. We will multiply this variable by 12 and take the log of it.
- Total number of children: Constructed using the same set of variables employed for the "daughter" variable. Capturing the total number of children a dyad has in a given survey year.
- Average weekly working hours and difference in working hours: Using the variable "plb0186_h" from the "pl" dataset, we will calculate the average number of weekly working hours reported by the couple. In addition, we will construct a measure of the difference in working time within the dyad. To create this variable, we will estimate the absolute difference in working hours between the two partners, divided by their total combined working hours. The resulting value (ranging from 0 to 1) will be subtracted from 1 (so that higher values represent a more equal distribution of working time between partners).

Note that some specifications will also include dyad and survey year fixed effects.

3 Inclusion/exclusion criteria

Our unit of analysis is the dyad (i.e., the couple) on a given survey year. The matching of individuals into couples will be performed using the variable "parid" from the "ppathl" dataset. We will employ all available data and do not restrict our sample to specific survey years. Besides this, we will apply the following inclusion criteria:

1. Couples that live together and unanimously identify as a couple:

   Using the variable "partner" from the "ppathl" dataset, we will retain only those individuals who (in a given year) either (1) report having a partner or (2) report having a spouse or registered partner (that is, individuals with a value of 1 or 2 in the "partner" variable). Based on how this "partner" variable is coded, this selection ensures that we only retain couples who (in a given year) reside in the same household, and who mutually identify each other as partners (i.e., the partnership is confirmed from both partners' perspectives).

2. Couples with complete data:

   For each hypothesis, we will only include observations with no missing values for any of the variables included in the relevant regression.

3. Couples with matching information:

   For each survey year, we will keep only those couples who report consistent information (i.e., identical values) on the following set of variables: the daughter binary indicator, the total number of children, the federal state indicators.

4. Couples with adult members:

   For each survey year, we will retain only those couples in which both partners are at least 18 years old.

5. Couples with realistic time-use reports:

   To reduce the influence of noisy data, for hypothesis 3, we will restrict our sample for a given year to couples in which both partners report spending fewer than 20 hours per day on housework and fewer than 20 hours per day on childcare (whether on weekdays, Saturdays, or Sundays). For each year, we will also focus exclusively on couples in which both partners report working less than 100 hours per week.

4 Exact analysis

In this section, we present the analysis for each hypothesis separately.

4.1 H1: Having daughters leads to stronger support for gender equality (as measured by attitudes).

To test this hypothesis, we will employ our attitudinal measure of support for gender equality as the dependent variable. Although this variable was collected across three different years, it was measured only once per individual. As a result, we have a single observation of the dependent variable per dyad (our unit of analysis). Given this data structure, we will simply estimate a cross-sectional linear OLS regression with robust standard errors. Specifically, we will estimate the following model:

$$Y_i = \beta_0 + \beta_1 \text{Daughter}_i + \mathbf{X}_i' \boldsymbol{\gamma} + \varepsilon_i \tag{1}$$

Where Y_i represents our dyadic attitudinal measure of support for gender equality by couple i (see "support for gender equality (attitudes)" in section 2), Daughter_i our main independent variable (representing whether couple i has at least one daughter), and $\mathbf{X}_{it}'$ a vector of demographic controls. These demographic controls include the average age of the couple, its squared term, a set of binary variables representing the employment status of the female partner, the employment status of the male partner, the highest school-leaving degree of the female partner, and the highest school-leaving degree of the male partner. We also include federal state indicators, the logarithm of yearly net household income, the total number of children reported by the couple, and survey year indicators.

4.2 H2: Having daughters leads to more left-leaning political preferences.

To test this hypothesis, we will employ our measure of left-leaning political preferences as the dependent variable. Since we have repeated measurements per dyad over time, we will estimate a linear model with dyad and survey year fixed effects. Specifically, we will estimate the following model using OLS with standard errors clustered at the dyad level:

$$P_{it} = \beta_0 + \beta_1 \text{Daughter}_{it} + \mathbf{X}_{it}' \boldsymbol{\gamma} + \alpha_i + \delta_t + \varepsilon_{it} \tag{2}$$

Where P_{it} represents the average political preference of the couple i in year t (see "left-leaning political preferences" in section 2), Daughter_i our main independent variable (representing whether couple i has at least one daughter in the year t, α_i and δ_t represent dyad and survey year fixed effects, respectively, and $\mathbf{X}_{it}'$ a vector of demographic controls. These demographic controls include the average age of the couple, its squared term, a set of binary variables representing the employment status of the female partner, and the employment status of the male partner. We also include the logarithm of yearly net household income, and the total number of children reported by the couple.

4.3 H3: Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

To test this hypothesis, we will employ our behavioral measure of gender equality as the dependent variable (see "gender equality (behaviors)" in section 2). Again, we have repeated measurements per dyad over time. Therefore, we will estimate a linear model with dyad and survey year fixed effects. Specifically, we will estimate the following model using OLS with standard errors clustered at the dyad level:

$$B_{it} = \beta_0 + \beta_1 \text{Daughter}_{it} + \mathbf{X}'_{it}\boldsymbol{\gamma} + \alpha_i + \delta_t + \varepsilon_{it} \tag{3}$$

Where B_{it} represents our behavioral measure of gender equality for couple i in year t (see "gender equality (behaviors)" in section 2), Daughter_i our main independent variable (representing whether couple i has at least one daughter in the year t, α_i and δ_t represent dyad and survey year fixed effects, respectively, and $\mathbf{X}'_{it}$ a vector of demographic controls. These demographic controls include the average age of the couple, its squared term, a set of binary variables representing the employment status of the female partner, and the employment status of the male partner. We also include the logarithm of yearly net household income, the total number of children reported by the couple, and the average and the difference in weekly working hours of the couple.

4.4 H4: Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

To test this hypothesis, we will employ our attitudinal measure of differences (within the couple) in support for gender equality as the dependent variable (see "differences in support for gender equality (attitudes)" in section 2). As in the first hypothesis, we have a single observation of the dependent variable per dyad (our unit of analysis). Given this data structure, we will simply estimate a cross-sectional linear OLS regression with robust standard errors. Specifically, we will estimate the following model:

$$D_i = \beta_0 + \beta_1 \text{Daughter}_i + \mathbf{X}'_i\boldsymbol{\gamma} + \varepsilon_i \tag{4}$$

Where D_i represents our dyadic attitudinal measure of differences in support for gender equality between male and female (see "differences in support for gender equality (attitudes)" in section 2), Daughter_i our main independent variable (representing whether couple i has at least one daughter), and $\mathbf{X}'_{it}$ a vector of demographic controls. These demographic controls include the average age of the couple, its squared term, a set of binary variables representing the employment status of the female partner, the employment status of the male partner, the highest school-leaving degree of the female partner, and the highest school-leaving degree of the male partner. We also include federal state indicators, the logarithm of yearly net household income, the total number of children reported by the couple, and survey year indicators.

5 Interpretation of results

For each hypothesis, our main test statistic will be the estimated coefficient on the variable "daughter".

For Hypotheses 1 through 3, we will interpret a positive and statistically significant coefficient (at the 95% confidence level) as evidence in support of the corresponding hypothesis.

Hypothesis 4 builds on the result of Hypothesis 1. If Hypothesis 1 is supported, and the coefficient on "daughter" in Model 4 is also positive and statistically significant at the 95% confidence level, we will interpret this as evidence in support of Hypothesis 4.